Approximating the Score Function using Semi-Discrete Optimal Transport

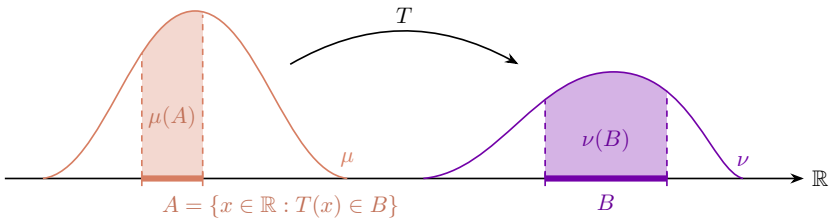
Curves and Surfaces 2026

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Laboratoire de Mathématiques d'Orsay

June 11th

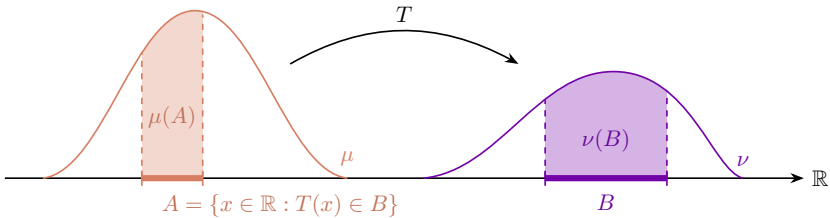
Quick reminders on Optimal Transport 1/2



- Given ground cost $c(x, y) = \|x - y\|^2$
- Among all maps T satisfying $\forall B,\ \nu(B) = \mu(T^{-1}(B))$
- Find the one minimising the average transport cost

$$W_2(\mu, \nu)^2 := \inf_{T\,:\,T_{\#}\mu = \nu} \int_{\mathbb{R}^d} \|x - T(x)\|^2 \mu(\mathrm{d}x)$$

Quick reminders on Optimal Transport 2/2

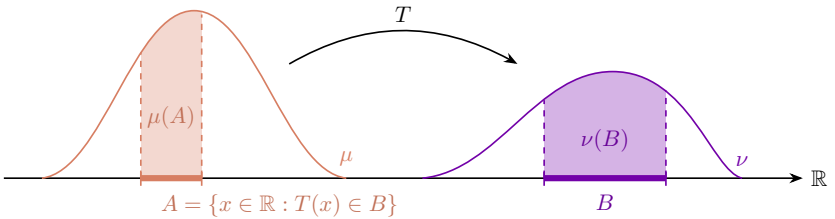


- Kantorovich duality

$$\inf_{T\,:\,T_{\#}\mu=\nu} \int_{\mathbb{R}^d} \|x-y\|^2 \mu(\mathrm{d}x) = \sup_{\varphi} \int \varphi(x)\mu(\mathrm{d}x) + \int \varphi^c(y)\nu(\mathrm{d}y),$$

where $\varphi^c(y) := \inf_x c(x,y) - \varphi(x)$.

Quick reminders on Optimal Transport 2/2



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- For $c(x,y) = \alpha\|x-y\|^2$

$$\nabla_x c(x, T^\star(x)) = \nabla\varphi^\star(x) \iff T^\star(x) - x = \boxed{y - x = -\frac{1}{2\alpha}\nabla\varphi^\star(x)}$$

Displacement

What is the score function?

- **Forward process:** Ornstein–Uhlenbeck (OU) process:

$$\mathrm{d}X_t = -X_t \mathrm{d}t + \sqrt{2}\mathrm{d}B_t, \quad X_0 \sim \mu, \quad t \in [0, T].$$

Let $\rho_t := \mathrm{Law}(X_t)$. Then ρ_t solves the Fokker–Planck equation

$$\partial_t \rho_t(x) = \Delta \rho_t(x) + \mathrm{div}(\rho_t(x)x). \tag{$\star$}$$

whose stationary solution is $\gamma \propto e^{-\|\cdot\|^2/2}$.

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- **Macroscopic POV:** Equation ($\star$) is the 2-Wasserstein Gradient Flow of the relative entropy $\mathcal{H}(\rho|\gamma) := \int \frac{\|x\|^2}{2}\rho + \int \rho \ln(\rho)$. It means ($\star$) can be written as

$$\partial_t \rho_t(x) = -\nabla \cdot (\rho_t(x) v_t(x)),$$

where $v_t(x) = -x - \nabla \ln(\rho_t(x))$.

Using OT to learn the score 1/2

- Given a time-step $\tau > 0$, such a gradient flow can be discretised in time using the **JKO scheme** (or Implicit Euler scheme, or Proximal step)

$$\begin{cases} \rho_0 = \mu \\ \rho_{n+1}^\tau := \arg\min_\rho \ \mathcal{H}(\rho|\gamma) + \dfrac{1}{2\tau} W_2(\rho_n^\tau, \rho)^2 \end{cases}$$

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- **At step n:** We show that

$$\rho_{n+1}^\tau \propto \exp\left(-(\varphi^\star)^{c/2\tau}\right) \mathrm{d}\gamma$$

where $\varphi^\star$ is a Kantorovich potential satisfying

$$\varphi^\star \in \arg\min_\varphi \ \tilde{J}(\varphi) := \ln\left(\int e^{-\varphi^{c/2\tau}} \mathrm{d}\gamma\right) - \int \varphi \mathrm{d}\rho_n^\tau.$$

Using OT to learn the score 2/2

- The optimal potential $\varphi^\star$ satisfies ($\alpha = \frac{1}{2\tau}$ here)

$$\boxed{-\tau \nabla \varphi^\star(x) = y - x}$$

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- The displacement of the particle x being $\tau v_t(x)$, we obtain

$$-\tau \nabla \varphi^\star(x) = \tau v_t(x) = -\tau(x + \nabla \ln\left(\rho^\tau_{n+1}(x)\right)) \iff \boxed{\nabla \ln\left(\rho^\tau_{n+1}(x)\right) = \nabla \varphi^\star(x) - x}.$$

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- We'll approximate the score by approximating the Kantorovich potential $\varphi^\star$, that minimises the functional:

$$\tilde{J}(\varphi) = \ln\left(\int e^{-\varphi^c/2\tau}\right) \mathrm{d}\gamma - \int \varphi \mathrm{d}\rho^\tau_n$$

General contributions

Proposition *(G.)*

The functional $\tilde{J}(\varphi) := \ln\left(\int e^{-\varphi^{c/2\tau}} \mathrm{d}\gamma\right) - \int \varphi \mathrm{d}\rho_n^\tau$ is convex.

General contributions

Proposition *(G.)*

The functional $\tilde{J}(\varphi) := \ln\left(\int e^{-\varphi^{c/2\tau}} \mathrm{d}\gamma\right) - \int \varphi \mathrm{d}\rho_n^\tau$ is convex.

Theorem *(G.)*

Let $J(\varphi) := \int e^{-\varphi^{c/2\tau}} \mathrm{d}\gamma - \int \varphi \mathrm{d}\rho_n^\tau$. Then,

$$\arg\min_\varphi J(\varphi) \subsetneq \arg\min_\varphi \tilde{J}(\varphi) \quad \text{and} \quad \arg\min_{\int e^{-\varphi^{c/2\tau}} \mathrm{d}\gamma = 1} \tilde{J}(\varphi) = \arg\min_\varphi J(\varphi).$$

Pro: J is "SGDisable"

Con: one more parameter to estimate (the normalisation constant!)

Stochastic gradient descent 1/3

- $J(\varphi) = \mathbb{E}_{y\sim\gamma}\left[\exp\left(-\varphi^{c/2\tau}(y)\right) - \int \varphi(x)\rho_n^{\tau}(\mathrm{d}x)\right]$

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- Choice of parametrisation: semi-discrete optimal transport using

$$\hat{\rho}_n^\tau = \frac{1}{N}\sum_{i=1}^N \delta_{\hat{x}_i^{(n)}} \rightsquigarrow \begin{cases} \varphi(\hat{x}_i^{(n)}) = \theta_i \\ \varphi^{c/2\tau}(y) = \min_i \left\{ \frac{c(\hat{x}_i^{(n)}, y)}{2\tau} - \theta_i \right\} \end{cases}$$

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- $G(\theta) := J(\varphi(\cdot\,;\theta)) = \mathbb{E}_{y\sim\gamma}\left[\exp\left(-\min_{1\le i\le N}\left\{\frac{c(\hat{x}_i^{(n)}, y)}{2\tau} - \theta_i\right\}\right) - \frac{1}{N}\sum_{i=1}^N \theta_i\right]$

Stochastic gradient descent 2/3

Projected SGD algorithm ($c_\tau(x,y) := \frac{c(x,y)}{2\tau}$) for some R_{proj}:

$$\begin{cases} \text{pick } y \sim \gamma \\ \text{let } i := \arg\min_j \; c_\tau(\hat{x}_j^{(n)}, y) - \theta_j \\ \text{define } e_i \text{ as the } i\text{-th canonical basis vector of } \mathbb{R}^N \\ \theta^{(k+1)} := \text{proj}_{B(0,R_{\text{proj}})} \left(\theta^{(k)} - \alpha_k \underbrace{e^{\theta_i^{(k)} - c_\tau(\hat{x}_j^{(n)}, y)}}_{\text{A priori } \textit{unbounded}!} e_i + \frac{\alpha_k}{N} \mathbb{1} \right) \end{cases}$$

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- Gradient bounded by $e^{R_{\text{proj}}}$

Stochastic gradient descent 3/3

Theorem *(G.)*

- $\alpha_k = \frac{C}{\sqrt{k}}, \quad \hat{\theta}_K := \frac{1}{K} \sum_{k=0}^{K} \theta^{(k)}$

Then, it holds that

$$0 \leq \mathbb{E}[G(\hat{\theta}_K)] - G(\theta^\star) \leq \frac{2R_{\text{proj}}^2}{CK} + \frac{B^2 C}{\sqrt{K}} = \mathcal{O}(K^{-\frac{1}{2}}),$$

where $\theta^\star$ denotes the unique minimiser of G.

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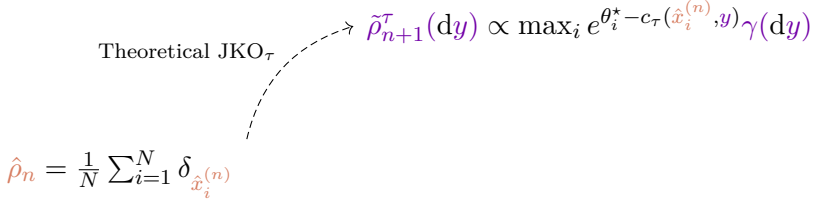
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- ☹

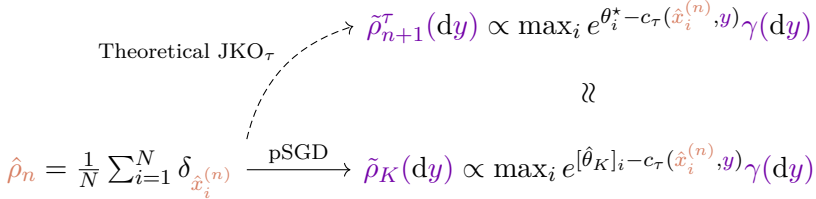
Approximating the next JKO iterate

$$\hat{\rho}_n = \frac{1}{N} \sum_{i=1}^{N} \delta_{\hat{x}_i^{(n)}}$$

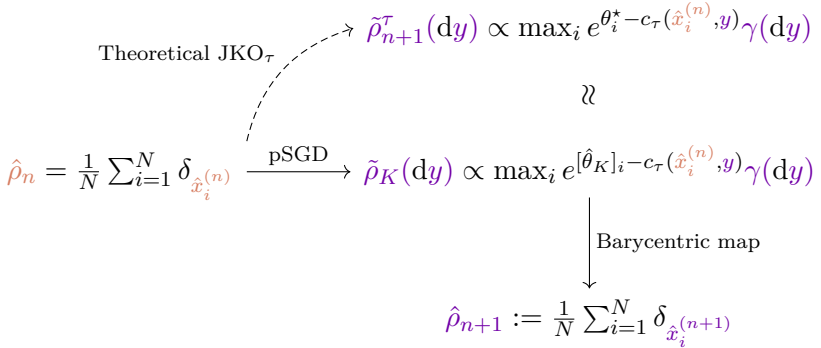
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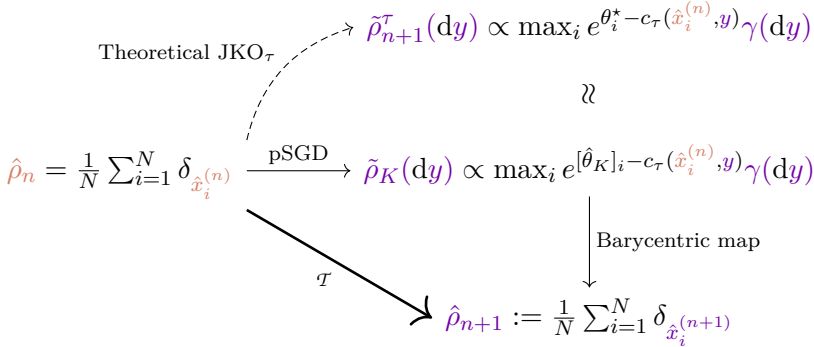
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More on the barycentric map

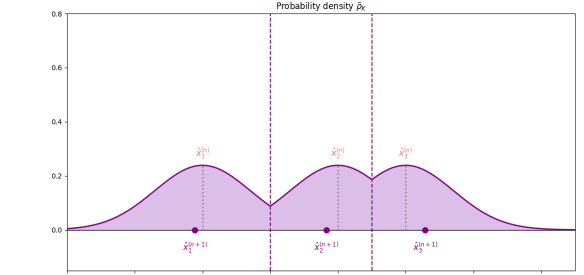
- $\rho(\mathrm{d}y) \propto \max_i e^{\theta_i - c_\tau(\hat{x}_i^{(n)}, y)} \gamma(\mathrm{d}y)$ is **piecewise log-quadratic**

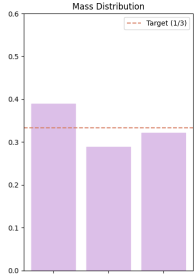
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  $L_i(\theta) := \left\{ y \in \mathbb{R}^d \ : \ \forall 1 \leq i \leq N,\ c_\tau(\hat{x}_i^{(n)}, y) + \theta_i \leq c_\tau(\hat{x}_j^{(n)}, y) + \theta_j \right\}$

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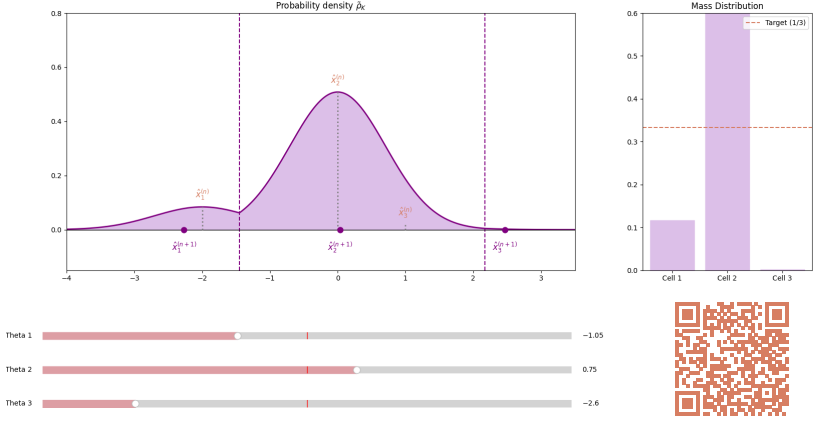






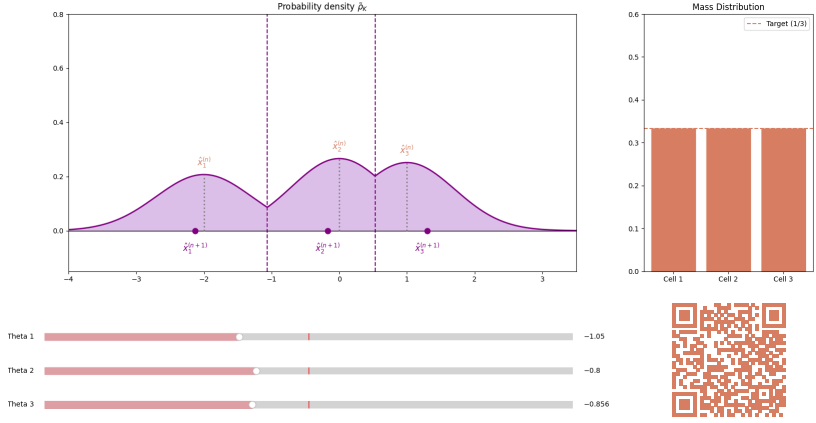
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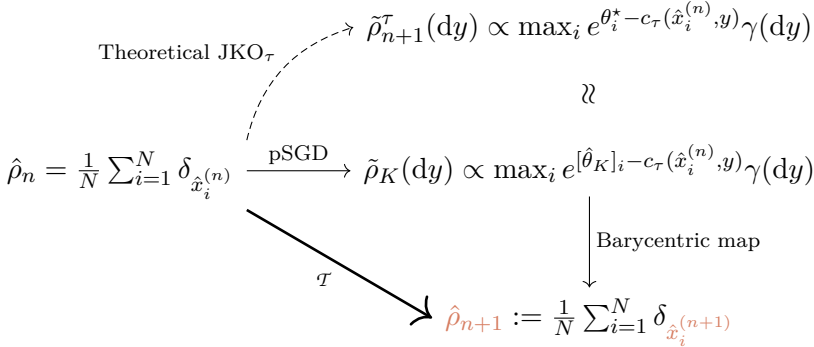


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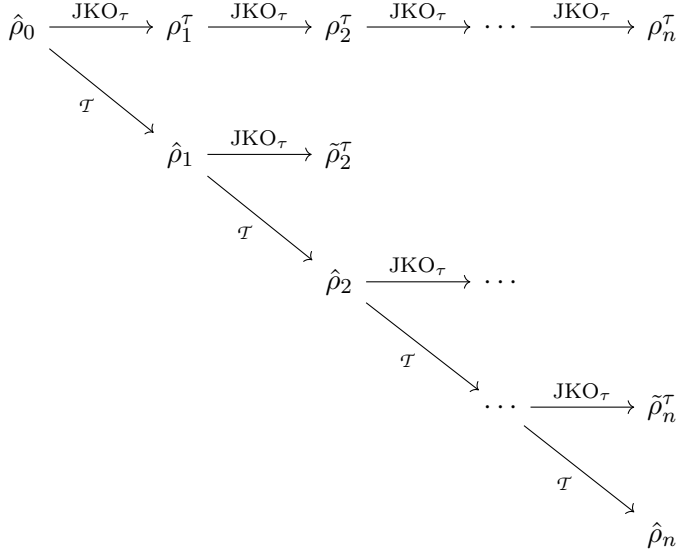
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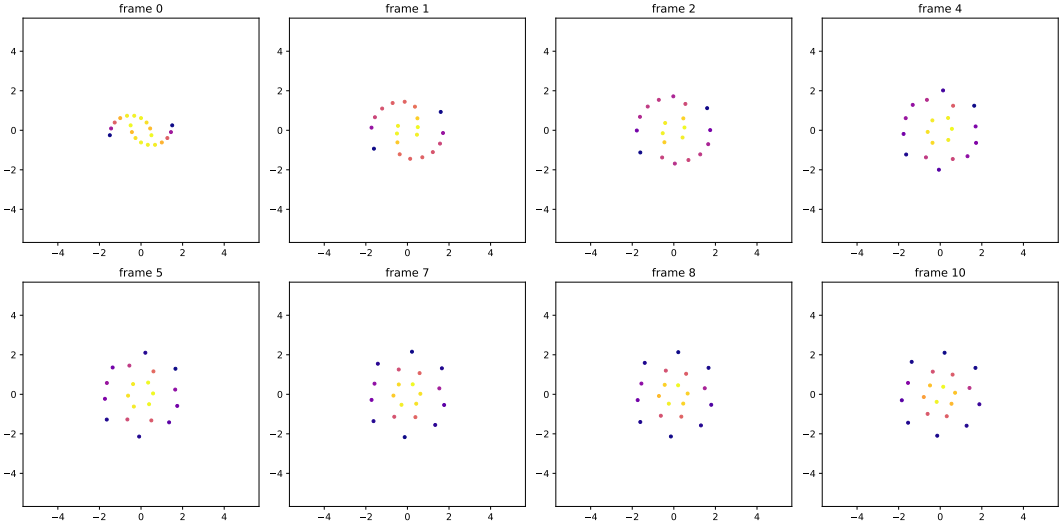
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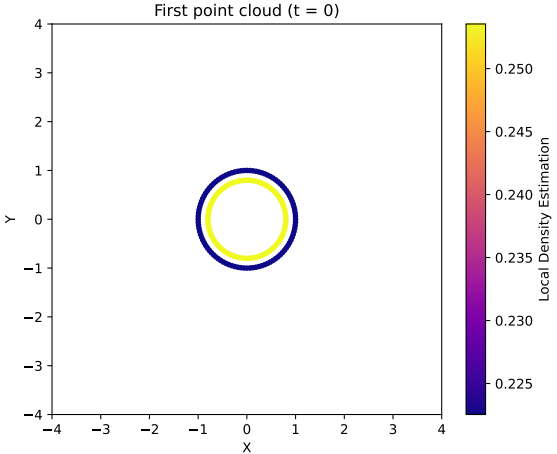
Iterating the whole procedure

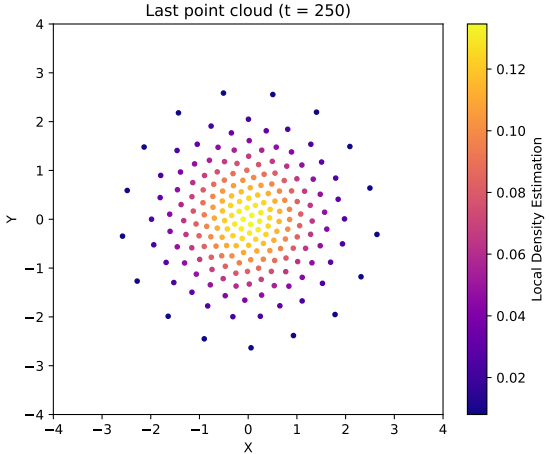


Some “animations”

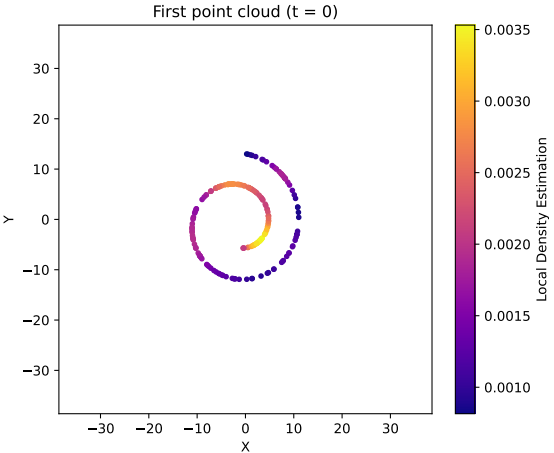


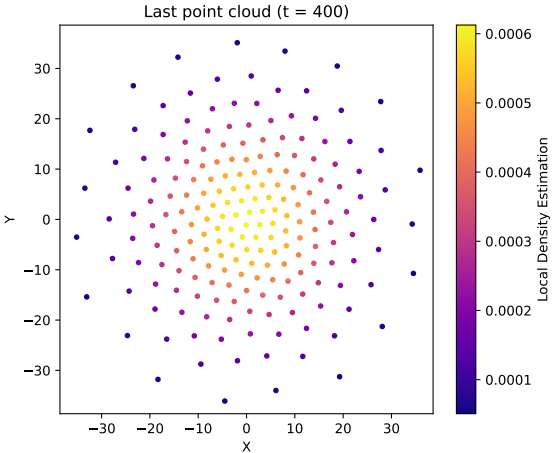
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Thank you for your attention!